

XIAORAN ZHA

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💻 Portfolio: <https://xrguin.github.io/>

🏠 Notre Dame, IN

Education

University of Notre Dame

Doctor of Philosophy in Electrical Engineering

Aug 2023 - Jun 2028 (Expected)

GPA: 3.7/4.0

University of California San Diego

Master of Science in Mechanical Engineering

Sep 2021 - Jun 2023

GPA: 3.6/4.0

The Ohio State University

Bachelor of Science in Mechanical Engineering

Aug 2017 - May 2021

GPA: 3.5/4.0

With Honors Research Distinction in Mechanical Engineering

Technical Skills

Programming & Machine learning: Python, PyTorch, MATLAB, C++

Control: MPC, PID, data-driven modeling and control, reinforcement learning, system identification

Robotics & Tools: ROS, OpenCV, Simulink, Linux, Arduino, Isaac Sim

Research Experience

University of Notre Dame - Robotics and Autonomy Research Lab

Aug 2023 - Current

Graduate Research Assistant, Mentor: Dr. Mengxue Hou

Notre Dame, IN

- Developed and deployed memory-augmented Koopman MPC for an underwater robot with online adaptation to address sim-to-real gap, reducing mean figure-eight lap time by 31% versus a baseline.
- Developed a Deep Koopman-based MPC framework for power grid voltage distribution control, achieving tracking performance equivalent to nonlinear MPC with average computation times of 0.07 s versus 28.88 s per update.
- Designed obstacle avoidance algorithms for multi-agent systems using time-series prediction models, achieving a 50% reduction in collision rate.
- Developed an OpenCV-based visual localization system for underwater robots using live smartphone video streams.

Oak Ridge National Laboratory

Jun 2024 - Aug 2024

Graduate Research Program (GRO), Mentor: Dr. Hong Wang, Dr. Yonghao Gui

Oak Ridge, TN

- Extended a Simulink power-grid model to incorporate environmental disturbances and developed a multilayer perceptron (MLP) model to predict grid-frequency probability distributions.
- Designed an optimization-based controller using the learned PDF model to select control inputs and track target frequency distributions.

University of California San Diego - Bioinspired Robotics and Design Lab

Mar 2022 - Jun 2023

Graduate Researcher, Mentor: Dr. Michael Tolley

San Diego, CA

- Developed scalable manufacturing methods for soft sensors using TPU sheets and heat-resistant mylar.
- Designed and implemented a real-time detection algorithm to identify activated sensor locations within a 3×3 sensor array using 6 transducers only.

The Ohio State University - Hydro and Aero Energy Lab

May 2020 - May 2021

Honors Undergraduate Researcher, Mentor: Dr. Clarissa Belloni

Columbus, OH

- Conducted CFD simulations of hydro-kinetic turbines using ANSYS Fluent on high-performance computing systems.
- Adjusted simulation parameters by comparing on-site collected data to reduce the sim-to-real gap, improving model accuracy and reliability.
- **HONOR:** Undergraduate Research Forum, 3rd place; Undergraduate Research Scholarship (\$2500)

Work Experience

Trina Solar

Apr 2023 - Jun 2023

Mechanical Engineering Intern

San Diego, CA

- Conducted mechanical analysis of battery storage racks to ensure they comply with seismic standards.
- Utilized SolidWorks to design and conduct finite element analysis on the structures of the energy storage system.
- Collaborated with other engineers to ensure the structure design meets all relevant safety and regulatory requirements.

SAIC Volkswagen Automotive Co., Ltd. - Car Body Engineering Dept.

Jun - Jul 2019

Automotive Engineer Intern

Shanghai, China

- Built the 3D model of EV (electric vehicle) battery packs with CATIA and contributed to a lighter-weight battery design.
- Coordinated with supplier partners, verified supplier components using CATIA and Excel, and yielded reports with data collection and frequent quality assurance checks.

Publications

- **Zha, X.**, Hou, M. (2025), A Mori–Zwanzig Formalism based Estimation Approach for Dynamic Obstacle Avoidance, 2025 22nd International Conference on Ubiquitous Robots
- **Zha, X.**, Jadhav, S., Adibnazari, I., Zhai, Y., & Tolley, M. T. (2024), Scalable Arrays of Pneumatic Sensors for Multitouch Soft Skins, Accepted by 2024 7th IEEE International Conference on Soft Robotics
- Y. Gui, H. Wang, **X. Zha** and Y. Xue, "Probability Density Function Control of Frequency Fluctuations in Renewable-Rich Power Systems," in IEEE Transactions on Sustainable Energy, doi: 10.1109/TSTE.2025.3578278.

Professional Service

- **Journal Reviewer:** Journal of Intelligent & Robotic Systems, IEEE Robotics and Automation Letters, IEEE Transactions on Automation Science and Engineering.
- **Conference Reviewer:** IROS, ICRA, CDC, ACC, UR.